MDRIIA Capstone: LinkedIn & GitHub Showcase Guide

Modern Day Robotics & Its Industrial Applications (MDRIIA - 702CO0E012) · B.Tech CSBS · AY 2026–27 Odd Semester · Dr. Sunny Nanade, MPSTME Mumbai

1. What you submit

#	Deliverable	Where	Marks it feeds
1	Your group's **Technical Dossier** (pre-filled Word file), completed and exported to **PDF**	MS Teams: **every student turns it in**. Also commit the filled `.docx` and `.pdf` to `Group_XX/docs/`	Component 1 CDIO Robotics & MuJoCo (10) and Component 2 CSBS Techno-Economics (5)
2	Your group folder in the course **monorepo**: model, controller, business script, real-trial CSVs	https://github.com/sunny-nanade/NMIMS-MPSTME-MDRIIA-2026	Components 1 and 2
3	A **3-to-4-minute LinkedIn video** + the live Expo demo and viva	LinkedIn post URL in dossier Section 5	Component 3 Showcase & Viva (5)

Due: Monday, 26-Oct-2026, 23:59 IST. Studio Showcase, Expo & Viva Voce: Saturday, 31-Oct-2026. No PPT slides. You present live software: the MuJoCo simulation running in its viewer, your business sensitivity graphs, and a webcam picture-in-picture of every member.

2. Filling in the dossier

- Open it in **Microsoft Word** (desktop). Every answer goes in a **"Select ▾" drop-down** or a **"type here" field**. Do not delete them, and do not edit the file in Google Docs: the fields are how your answers are read.
- Pictures (FBD, plots, code) go in the boxed areas. Paste code as text, not as a screenshot.
- **Section 6** is not marked. Each member fills their own row honestly. Research consent is optional and does not affect marks.
- Save, then **File → Export → PDF**. Commit both files to `Group_XX/docs/` with the original name `MDRIIA_Group_XX_Technical_Dossier`.

3. The individual turn-in rule

Marks are entered **per student** in the NMIMS portal. **Every one of you must attach the group's final PDF on MS Teams and click "Turn in" yourself.** No turn-in, no marks recorded, even if your teammates submitted.

4. GitHub: the course monorepo

- All 12 groups work in one repository. Your work lives only in your folder: `models/` (MJCF), `src/` (controller), `analytics/` (business script, real-trial CSVs), `docs/` (dossier, figures).
- **Each member commits their own work from their own GitHub account.** Section 6 asks for your commit share; it is checked against `git shortlog -sn -- Group_XX_*/`.
- Work on a branch (e.g. `group-07/grasp-fsm`) and open a pull request into `main`. Pull before you push.
- Do not commit videos, virtual environments (`venv/`) or large binaries. Put the video on LinkedIn.

5. Data integrity: simulated targets vs your real trials

- Files named `**SIMULATED_*` **and every `generate_paper_figures.py` produce** randomly generated target numbers **(`np.random`). They show the CSV format and the plot style.** They are not results.**
- Your results come only from **MuJoCo runs of your own controller**, logged by your code to `analytics/*_real_trials.csv` (N ≥ 30 runs per condition, 500 Hz, seeds recorded).
- Every economic input needs a source, or must be labelled an assumption and varied in your sensitivity graph. Hard-coded "results" in a script (e.g. `payback_months = 13.4`) must be replaced by calculations.
- Reporting generated numbers as results is research misconduct and scores zero for that component.

6. Engineering and business checklist

- **Physics:** the model loads (`mujoco.MjModel.from_xml_path`), the timestep is 0.002 s, contact friction and `solref` are justified, and every FSM state is visible in the viewer.
- **Control:** PID / impedance / DLS-IK gains stated with units; tracking RMSE and control effort $\int \tau^2 \, dt$ logged.
- **Statistics:** baseline vs proposed, N ≥ 30 each, Welch's t-test, Cohen's d, 95 % CI.
- **Business:** H_reclaimed (labour reallocation), dimensionless P_payback, NCRI (= 1 − κ), an M/M/1 or M/M/c queuing delay, and an FMEA with RPN before and after mitigation. No currency amounts in the title, abstract or video: use ratios.

7. The LinkedIn video (3:00–4:00, no slides)

Time	Part	On screen
0:00–0:20	The problem and your Aalborg question, with one verified number	Webcam
0:20–1:40	**Live MuJoCo simulation**: show the FSM states, the contact behaviour and one failure your controller handles	Viewer + webcam PiP
1:40–2:20	Results: N runs, RMSE, primary metric, the t-test	Plot + webcam PiP
2:20–3:00	**Business sensitivity graph**: payback, NCRI, queuing delay, top FMEA risk	Graph + webcam PiP
3:00–3:45	Each member: what I built, what I learned, one thing AI got wrong	Webcams

- **Every member appears and speaks.** A Teams, Zoom or Meet recording with screen share and webcams on is fine.
- Upload natively on LinkedIn (no YouTube link), one post per group, teammates tagged; tag MPSTME and Dr. Sunny Nanade; link your monorepo folder.
- **Privacy:** no SAP IDs, no phone numbers, no personal data of anyone outside the team.

Post template

```
[One-line hook with one verified number]
Our MDRIIA capstone team (B.Tech CSBS, SVKM's NMIMS MPSTME) asked: "[Aalborg title]"
We modelled [robot] in DeepMind MuJoCo with a [control law] controller and ran [N] Monte Carlo trials:
[primary metric result], RMSE [x]. Business case: payback [P], NCRI [x].
What we learned: [one honest line, including one thing AI got wrong]
Code: [link to your monorepo folder]   Team: @… @…   Guided by @Sunny Nanade
#Robotics #MuJoCo #CSBS #TCS #MPSTME #NMIMS
```

8. AI use and the audit log

AI tools are allowed; **unverified AI output is not.** Section 1.1 asks you to verify each carried-over claim (source type and Keep / Correct / Delete). Section 5.1 logs three cases where AI gave wrong robotics or economic data, how you caught each one (textbook, MuJoCo documentation, the standard, the original paper, or running the code) and what kind of error it was.

9. Peer review (Section 5.2)

Score two other groups' videos 0–2 on Problem, Physics, Business, Data and Clarity, with the best moment and one viva question.

Your group	You review
Group 01	Group 02 and Group 07
Group 02	Group 03 and Group 08
Group 03	Group 04 and Group 09
Group 04	Group 05 and Group 10
Group 05	Group 06 and Group 11
Group 06	Group 07 and Group 12
Group 07	Group 08 and Group 01
Group 08	Group 09 and Group 02
Group 09	Group 10 and Group 03
Group 10	Group 11 and Group 04
Group 11	Group 12 and Group 05
Group 12	Group 01 and Group 06

10. Timeline to the deadline

Date	Milestone
Fri 16-Oct	Packs released on MS Teams; open your dossier and check your roster
Tue 20-Oct	Last day to report roster corrections (name, roll, SAP ID, role)
Thu 22-Oct	Model loads, controller runs, 500 Hz logging to `*_real_trials.csv` works
Sat 24-Oct	N ≥ 30 Monte Carlo runs per condition done; business model, FMEA and sensitivity graph done
Sun 25-Oct, 18:00	**LinkedIn video posted** (peer reviewers need it before the deadline)
Mon 26-Oct, 23:59 IST	**Dossier PDF turned in by every member on MS Teams; filled .docx + .pdf committed**
Sat 31-Oct	Studio Showcase, Expo & Viva Voce (slots posted on MS Teams)

11. Prepare for these viva questions

- Why did you choose that `solref` time constant and friction? What happens in the viewer if you halve them?
- Show one FSM transition live and point to the line of code that triggers it.
- Which single input moves your payback horizon most, and where does its value come from?
- Why Welch's t-test? What does your Cohen's d mean in plain words?
- What did AI get wrong in your project, and how did you prove it?

12. Expo & viva voce (Saturday, 31-Oct-2026)

Bring a laptop with your simulation running from a fresh `git clone` and `pip install -r requirements.txt`. Each member will be asked about **their own role and code** and about the whole project, including the business model. If you cannot explain something you submitted, it counts against you. Optional Track A: your dossier maps onto a 4-page IEEE conference paper (see the end of Section 6 of your dossier); speak to Dr. Nanade before submitting anywhere.